Bias from Treating Competing Risks as Censoring

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Abstract

In competing risks settings, treating competing events as ordinary censoring can lead to biased estimates of absolute risk. This simulation study compares a naive Kaplan–Meier approach with the cumulative incidence function for estimating 5-year relapse probability when death before relapse acts as a competing event. Across 1000 simulated datasets per scenario, the cumulative incidence estimator recovered the analytical truth, while the naive Kaplan–Meier estimator increasingly overestimated relapse risk as the competing event rate increased. Under the strongest competing risk scenario, the relative overestimation was approximately 42%. These results illustrate that the problem is not merely numerical but estimand-related.

Introduction

Survival Analysis concerns time-to-event data. It is usual to observe a situation where the event of interest coexists with another event, called a "competing risk". Consider for instance the case where the event of our interest is the relapse after treatment, and death, when it happens before relapse, is considered as a "competing event". If our aim is to estimate the risk of the event of interest, then one problematic setting occurs when considering death as censoring and estimating through $1 - \widehat{S}(t)$, where $\widehat{S}(t)$ is the Kaplan-Meier estimator. This issue is well known in the competing risks literature, where cumulative incidence is distinguished from naive Kaplan-Meier-type estimation that treats competing events as ordinary censoring (Gooley et al., 1999; Putter et al., 2007). The aim of this note is to quantify the bias induced by considering competing events as censoring.

Simulation Design

For each simulated individual, two event times were generated

$T_1 \sim \mathrm{Exp}(\lambda_1)$, for relapse, and

$T_2 \sim \mathrm{Exp}(\lambda_2)$, for death before relapse.

The observed time for each individual was

$T = \min\{T_1, T_2\}$.

The event indicator was defined as

$$I = \begin{cases} 1, & \text{if } T_1 < T_2 \\ 2, & \text{otherwise.} \end{cases}$$

The relapse rate was fixed at

$$\lambda_1 = 0.08$$

while the competing risk rate varied across the following cases

$$\lambda_2 \in \{0, 0.04, 0.08, 0.16\}.$$

We also imposed administrative censoring (considered as the period of our study) at 5 years. Each scenario used $n = 1000$ individuals and we generated $B = 1000$ samples for each one (scenario).

True Cumulative Incidence Function

Under exponential competing risks, the true cumulative incidence of relapse is given by

$$F_1(t) = \frac{\lambda_1}{\lambda_1 + \lambda_2}\left(1 - \exp\{-(\lambda_1 + \lambda_2)\,t\}\right).$$

When $\lambda_2 = 0$ the above is reduced to

$$F_1(t) = 1 - \exp\{-\lambda_1 t\}.$$

What does the naive Kaplan-Meier estimator target?

In the competing risks setting, the cumulative incidence function estimates the real-world probability of relapse by time t. We've stated above the full formula with $F_1(t)$.

By contrast, if deaths before relapse are treated as censoring, the naive Kaplan-Meier estimator of the relapse probability is

$$1 - S_1(t) = 1 - \exp\left\{-\lambda_1 t\right\}.$$

This quantity does not depend on the event rate λ_2. It can be interpreted as a hypothetical relapse risk in a setting where the competing event is ignored or removed. Generally, it is not equal to the real-world cumulative incidence $F_1(t)$.

In fact, the difference between the naive Kaplan-Meier estimator and the cumulative incidence is

$$\Delta(t) = [1 - \exp(-\lambda_1 t)] - \frac{\lambda_1}{\lambda_1 + \lambda_2}\left[1 - \exp\{-(\lambda_1 + \lambda_2)t\}\right].$$

When $\lambda_2 = 0$, this difference is equal to 0. However, when $\lambda_2 > 0$, the naive Kaplan-Meier estimand is greater than the real-world cumulative incidence. Since naive Kaplan-Meier treats competing events as non-existent ($\lambda_2 = 0$), it is clear that its estimand remains approximately constant, although λ_2 increases and, consequently, cumulative incidence function decreases.

Comparison of the Methods

Two kinds of estimators for the 5-year risk of relapse were examined:

1) A naive Kaplan-Meier estimator, where deaths before relapse were treated as censored observations, and

2) The cumulative incidence function estimator, where death before relapse was treated as a competing risk.

For each method, bias was computed as

$$\text{bias} = \widehat{F} - F,$$

where $\widehat{F}$ is the estimation of the relapse risk and F is the true cumulative incidence.

Although this note does not fit regression models, the cumulative incidence framework is closely related to the subdistribution approach formalised by Fine and Gray (1999).

Results

Table 1: Simulation results for 5-year relapse probability estimation.

λ_2	True CIF	Naive KM	CIF	KM bias	CIF bias	KM relative bias (%)
0.00	0.3297	0.3296	0.3296	-0.0001	-1e-04	-0.02
0.04	0.3008	0.3301	0.3011	0.0293	3e-04	9.73
0.08	0.2753	0.3303	0.2759	0.0550	6e-04	19.96
0.16	0.2329	0.3302	0.2332	0.0973	3e-04	41.76

Across 1000 simulated datasets per scenario, the CIF estimator closely recovered the analytical 5-year relapse probability in all settings. When no competing event was present, the naive Kaplan-Meier estimator also agreed with the true cumulative incidence.

As the competing event rate increased, however, the naive Kaplan-Meier estimator increasingly overestimated the real-world 5-year relapse probability. For $\lambda_2 = 0.04$, the true 5-year relapse probability was 0.3008, while the naive Kaplan-Meier estimate was 0.3301, corresponding to an absolute overestimation of 0.0293. For $\lambda_2 = 0.08$, the overestimation increased to 0.0550. Under the strongest competing event scenario, $\lambda_2 = 0.16$, the true 5-year relapse probability was 0.2329, whereas the naive Kaplan-Meier estimate was 0.3302, corresponding to an absolute overestimation of 0.0973 and a relative overestimation of approximately 41.8%.

In contrast, the CIF estimator remained close to the analytical truth across all scenarios, with negligible bias.

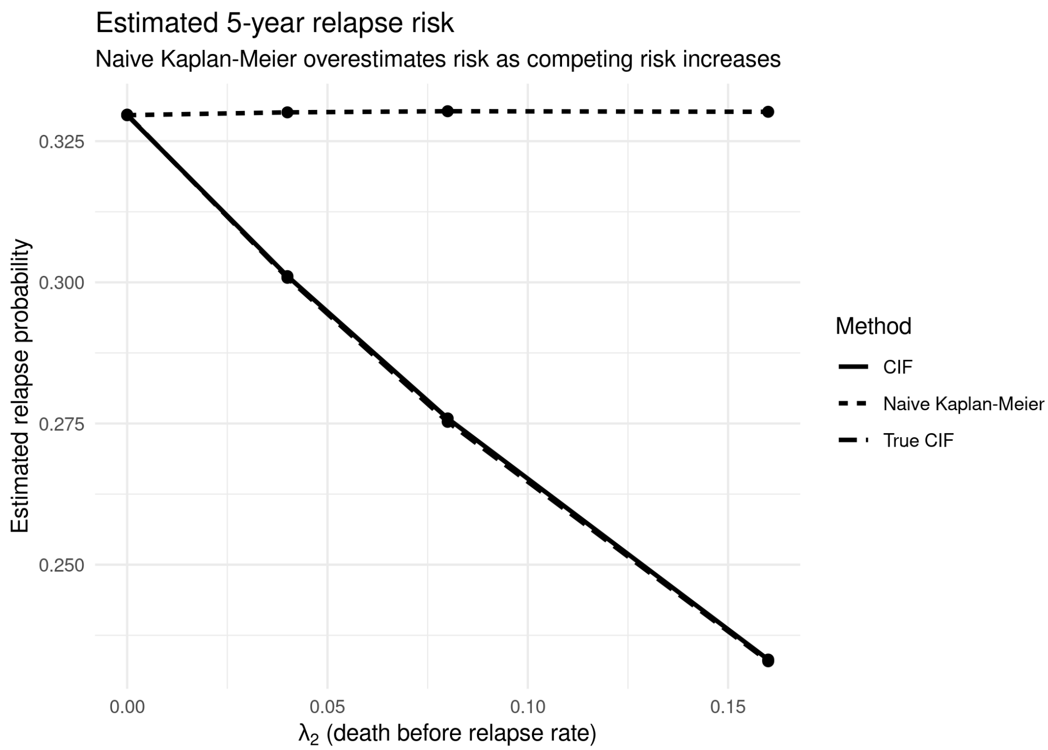

Figure 1: Estimated 5-year relapse probability under increasing competing event rates.

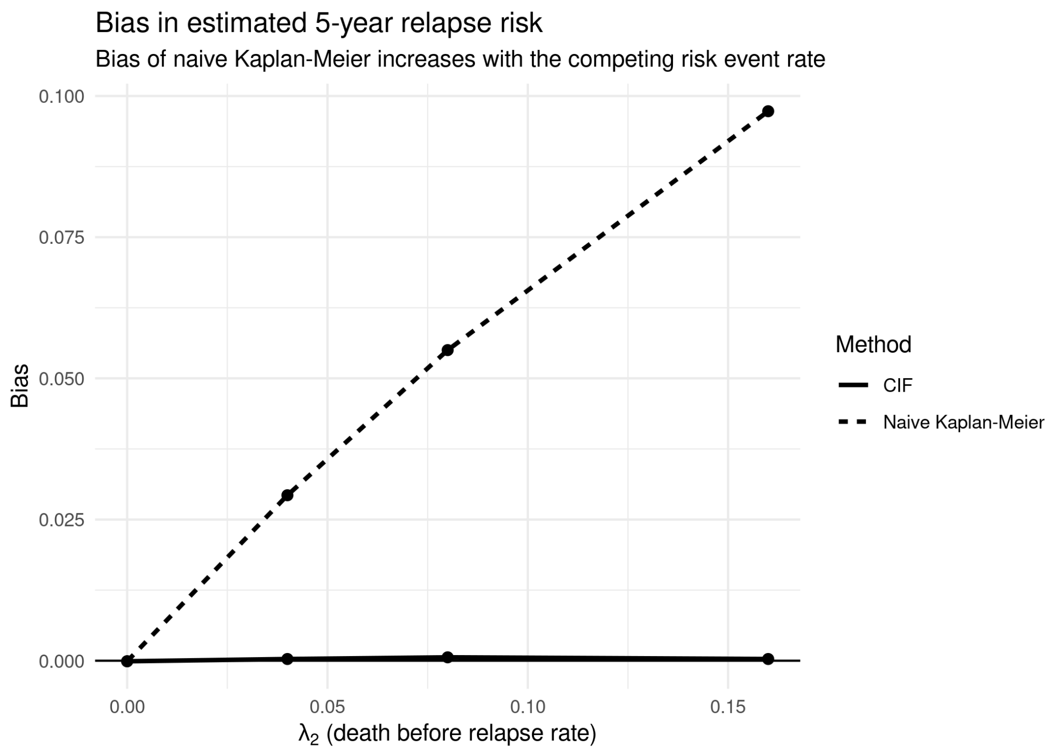

Figure 2: Bias in estimated 5-year relapse probability as the competing event rate increases.

Sample Size Sensitivity

We now examine the case where the sample size varies. Keeping λ_1 constant and $\lambda_2 \in \{0, 0.04, 0.08, 0.16\}$ we now check for different sample sizes $n \in \{250, 500, 1000, 2000\}$. The bias of the naive Kaplan-Meier estimator does not change even if the sample's variability becomes smaller. This happens since the naive Kaplan-Meier estimator aims for the estimand and this estimand is wrong if we consider dead individuals as censored.

Table 2: Sample size sensitivity analysis.

n	True CIF	Naive KM	CIF	KM bias	CIF bias	SD KM	SD CIF
250	0.3297	0.3314	0.3314	0.0017	0.0017	0.0304	0.0304
250	0.3008	0.3279	0.2989	0.0271	-0.0019	0.0324	0.0301
250	0.2753	0.3290	0.2748	0.0537	-0.0005	0.0341	0.0294
250	0.2329	0.3287	0.2323	0.0958	-0.0006	0.0374	0.0264
500	0.3297	0.3291	0.3291	-0.0006	-0.0006	0.0204	0.0204
500	0.3008	0.3298	0.3009	0.0290	0.0001	0.0223	0.0206

n	True CIF	Naive KM	CIF	KM bias	CIF bias	SD KM	SD CIF
500	0.2753	0.3291	0.2748	0.0537	-0.0005	0.0246	0.0212
500	0.2329	0.3306	0.2338	0.0976	0.0008	0.0256	0.0185
1000	0.3297	0.3295	0.3295	-0.0002	-0.0002	0.0149	0.0149
1000	0.3008	0.3303	0.3014	0.0295	0.0006	0.0159	0.0149
1000	0.2753	0.3302	0.2757	0.0549	0.0004	0.0167	0.0141
1000	0.2329	0.3301	0.2331	0.0971	0.0001	0.0187	0.0134
2000	0.3297	0.3301	0.3301	0.0004	0.0004	0.0107	0.0107
2000	0.3008	0.3297	0.3008	0.0289	0.0000	0.0113	0.0103
2000	0.2753	0.3293	0.2750	0.0539	-0.0004	0.0119	0.0101
2000	0.2329	0.3298	0.2331	0.0968	0.0002	0.0135	0.0097

The sample size sensitivity analysis shows that increasing (n) reduces the empirical variability of both estimators. However, the bias of the naive Kaplan-Meier estimator remains approximately unchanged. This indicates that the overestimation is not a finite-sample artifact. Rather, it is caused by the fact that the naive Kaplan-Meier estimator targets a hypothetical relapse risk that ignores the competing event.

Discussion

This simulation analysis illustrated a fundamental problem in assessing risks for an event, when a competing event coexists. The main problem is not just a numerical failure due to the use of a certain estimator. We observe that when the rate of the competing risk is increased, Kaplan-Meier estimator overestimates the risk of relapse before death, treating dead individuals (competing event) as censored data. On the other hand, cumulative incidence function estimator has small bias in estimating the probability of the event of interest, since it treats competing events as competing events and not as censored data. This overestimation can be clinically meaningful, especially in settings where competing events are common, such as oncology or heart failure studies.

Limitations

In this short note, we only examined data with independent exponential times. No covariates, no treatment groups, no random censoring, no non-constant hazards included. These simplifications were intentional in order to illustrate how naive Kaplan-Meier estimator fails to estimate accurately the risk of relapse when a competing event could happen before that. Future extensions could include covariate-dependent event rates, non-exponential survival times, random censoring, Fine-Gray models.

Conclusion

This simulation study shows that treating competing events as ordinary censoring can lead to substantial overestimation of real-world absolute risk. In this setting, the naive Kaplan-Meier estimator increasingly overestimated the 5-year relapse probability as the competing event rate increased, whereas the cumulative incidence function estimator recovered the analytical truth. The key issue is estimand-related: in competing risks settings, $1-\widehat{S}(t)$ after censoring competing events does not estimate the real-world cumulative incidence of the event of interest.

References

Gooley, T. A., Leisenring, W., Crowley, J., & Storer, B. E. (1999). Estimation of failure probabilities in the presence of competing risks: New representations of old estimators. *Statistics in Medicine*, 18(6), 695–706.

Putter, H., Fiocco, M., & Geskus, R. B. (2007). Tutorial in biostatistics: Competing risks and multi-state models. *Statistics in Medicine*, 26(11), 2389–2430.

Fine, J. P., & Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *Journal of the American Statistical Association*, 94(446), 496–509.